

Dr. Sai Srujan Kumar Chalavadi

Research Associate - III

Indian Institute of Technology, Roorkee | Website

S-206, Dept. of Earthquake Engg., IIT Roorkee, India +91 - 9629606785 srujanchalavadi6@gmail.com

EDUCATION

Program	Institution	%/CGPA	Year
Ph.D.	Indian Institute of Technology Roorkee, Roorkee (IITR)	**8.86**/10	2026
B Tech	National Institute of Technology Tiruchirappalli, Trichy (NITT)	**9.49**/10	2018
Class XII	Sri Chaitanya Junior College, Vijayawada	**97.9**/100	2014
Class X	S.T.B.E.M. High School, Vijayawada	**9.70**/10	2012

AREAS OF INTEREST

- Rock mechanics, Geotechnical earthquake engineering.
- Design, fragility and risk assessment of underground structures and slopes.

SKILLS

- **Numerical modelling**: MIDAS GTS NX 2026, RocScience Suite, UDEC ver. 7.0, Plaxis 3D 2025.
- **Seismic design of tunnels**: DARTS ver. 3.0.
- MATLAB 2026, Autodesk AutoCAD 2027, Microsoft 365 suite

PUBLICATIONS

- Chalavadi, S., Singh, Y. and Singh, M., 2026. Seismic fragility analysis of rock tunnels using multi-stripe analysis. *Rock Mechanics and Rock Engineering*. (Link)
- Chalavadi, S., Singh, Y. and Singh, M., 2026. Effect of topographic amplification on seismic vulnerability of mountain tunnels using multi-stripe analysis. *Rock Mechanics and Rock Engineering*. (Link)
- Chalavadi, S., Singh, Y. and Singh, M., 2026. Seismic behaviour of tunnels in Indian Himalayas. In: Deb, D., Murthy, V.M.S.R., Venkatesh, H.S., Rao, K.S., Goel, R.K. and Singh, M., Eds., *2024 ISRM International Symposium: ARMS13 - Advances in Rock Mechanics — Infrastructure Development, Lecture Notes in Civil Engineering*, Vol. 5. Springer, Cham. (Link)
- Chalavadi, S., Singh, M., Singh, Y. and Choudhari, J., 2023. Effect of scale and in-situ stress ratio on the deformation modulus of rock mass around tunnels. *Proceedings of the 15th ISRM Congress*, Salzburg, Austria, October 9–14, 2023. (Link)

SCHOLARSHIPS/AWARDS

- **International Society of Rock Mechanics Early Career Forum** fund to attend 13th Asian Rock Mechanics Symposium (ARMS13), one among 8 participants selected across Asia.

- **ISRM Best Paper Award**, for my research presented at ARMS13, New Delhi.
- **Prime Minister Research Fellowship** (PMRF) and a grant of **₹ one million**, funded by Ministry of Human Resource Development, Govt. of India.
- **Central Sector Scheme of Scholarship** (CSSS) funded by Ministry of Education, Govt. of India for outstanding performance in Class XII.

PROJECTS/INTERNSHIPS

- Seismic Vulnerability of Tunnels in Himalayas (Granted by **MoRTH, 9 million ₹, based on my thesis**)

  Supervisor: Prof. Yogendra Singh **University:** IIT Roorkee, India **Year:** 2025-Present

- Effect of high-speed train induced local liquefaction around piles in structures using PLAXIS 3D

  Supervisor: Prof. K. Muthukkumaran University: NIT Tiruchirappalli, India Year: 2018

- Effect of variation in unsaturated shear strength in the residual soil slopes across Singapore

  Supervisor: Prof. Harianto Rahardjo University: NTU, Singapore Year: 2017

- Studies on local site effects for a site located at Goa city using FLAC 2D

  Supervisor: Prof. A. Boominathan University: IIT Madras, India Year: 2016

WORK EXPERIENCE

- **Doctor of Philosophy** (On-site/Full-time) (July 2018-June 2026)
  - Thesis title: Seismic Fragility Analysis of Rock Tunnels using Multi-Stripe Method
  - Areas of research: Rock mechanics, Underground structures (tunnels), IS code-based seismic tunnel design, slope stability analysis, Hazard assessment, Vulnerability modelling, Performance-based design.
- **AI Model Trainer at Poindexter Labs** (Remote/Contract) (Sept – Nov 2025)
  - Served as Contributor and Reviewer across three projects (Canary, Hippo, and Phoenix) that specializes in creating graduate and post-graduate level prompts to train AI models.
- **AI Model Trainer at Outlier** (Remote/Contract) (May – July 2025)
  - Served as Contributor, Reviewer, and Senior Reviewer across two projects, consistently receiving high performance ratings (average 5.0 and 4.2).
- **Teaching Assistant** (On-site/Part-time) (May 2020-Present)
  - Lead tutorial sessions for graduate and postgraduate students, including facilitating discussions, resolving academic queries, and delivering supplementary lectures as needed.

MANAGEMENT POSITIONS

- **Sports secretary** at Bhawan Council, Azad Bhawan, IIT Roorkee. 2023-24
- **Event organizer** for one day seminar on Landslide Analysis and Mitigation (LAM), IIT Roorkee. 2018
- **President** at Builders' Hive – Civil Engineering Research & Development club of NIT Trichy. 2017-18

- **Event Coordinator** at Pragyan **–** A International Techno-management festival of NIT Trichy**.** 2015-16
- **Events coordinator** at Moments **–** A National level technical symposium of CED, NIT Trichy 2015-18

ALL ACADEMIC ACHIEVEMENTS

- Our team reached semi-finals at **Rock Bowl** event in **ISRM 2023** at Salzburg, Austria, one among 21 teams selected across the world.
- Selected for international school on LAndslide Risk Assessment and Mitigation (LARAM) in 2020, organized by University of Salerno.
- Successfully presented our team project "**Modified Falkirk Wheel**" at SANGAM project exhibition of **Pragyan'16 –** An International Techno-Management fest of NIT Trichy.
- Participated in **Mortar Master** - an event at **Pragyan'16** - An International Techno-Management fest of NIT Trichy.
- Won *1st prize* in **Bridge it** - an event at **Thatva'15**- A National level Technical fest of NIT Calicut.
- Won *1st prize* in **Bridge Modelling** - an event at **Rickem'15**- A National level technical symposium of King's college.
- Poster Presentation on **Seismic Retrofitting in Structures**- at **Rickem'15**- A National level technical symposium of King's college
- Won *3rd prize* in a **Poster presentation** on Self-cleaning concrete- an event at **Vitruvius'15**- A National level technical symposium of M.A.M College.
- Secured **best outgoing student** award at the school level.

LANGUAGES

- English: Fluent (Professional working proficiency)
- Hindi: Native/Fluent
- Telugu: Native/Fluent

MEMBERSHIPS

- International Society for Rock Mechanics and Rock Engineering (Through Indian Committee)
- International Society for Soil Mechanics and Geotechnical Engineering (Through Indian Committee)
- Indian Geotechnical Society (LM-4577)
- Indian Society of Earthquake Technology (LM-1852)
- Indian Society for Rock Mechanics and Tunnelling Technology (LM-2280)
- Society for Earthquake and Civil Engineering Dynamics

EXTRA-CURRICULAR ACTIVITIES

- Received multiple trophies at Bhawan and Institute level in Badminton tournaments at IIT Roorkee.
- Over the years I have developed a passion for Touch Typing. My current speed stands at 110 WPM.
- Passionate about learning Chess and playing Flute.